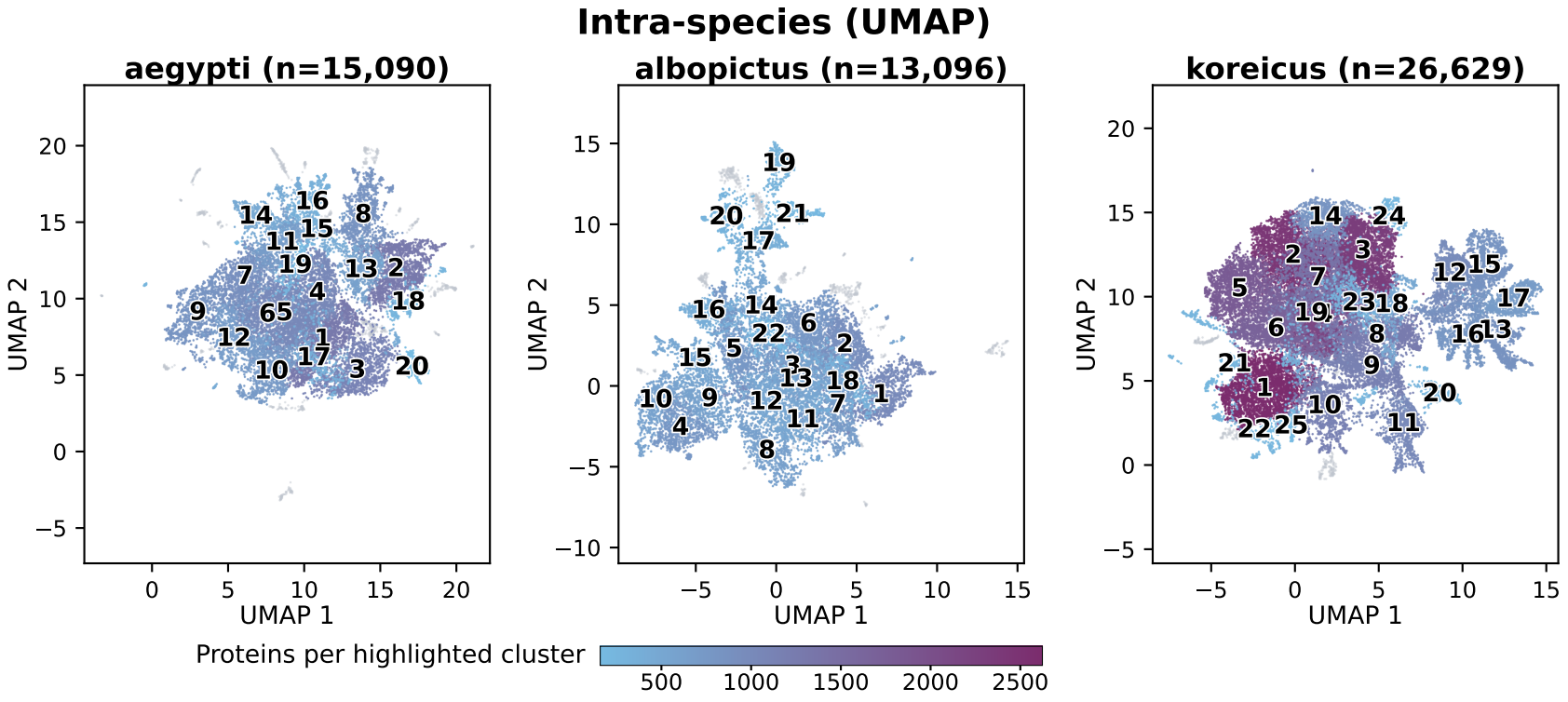
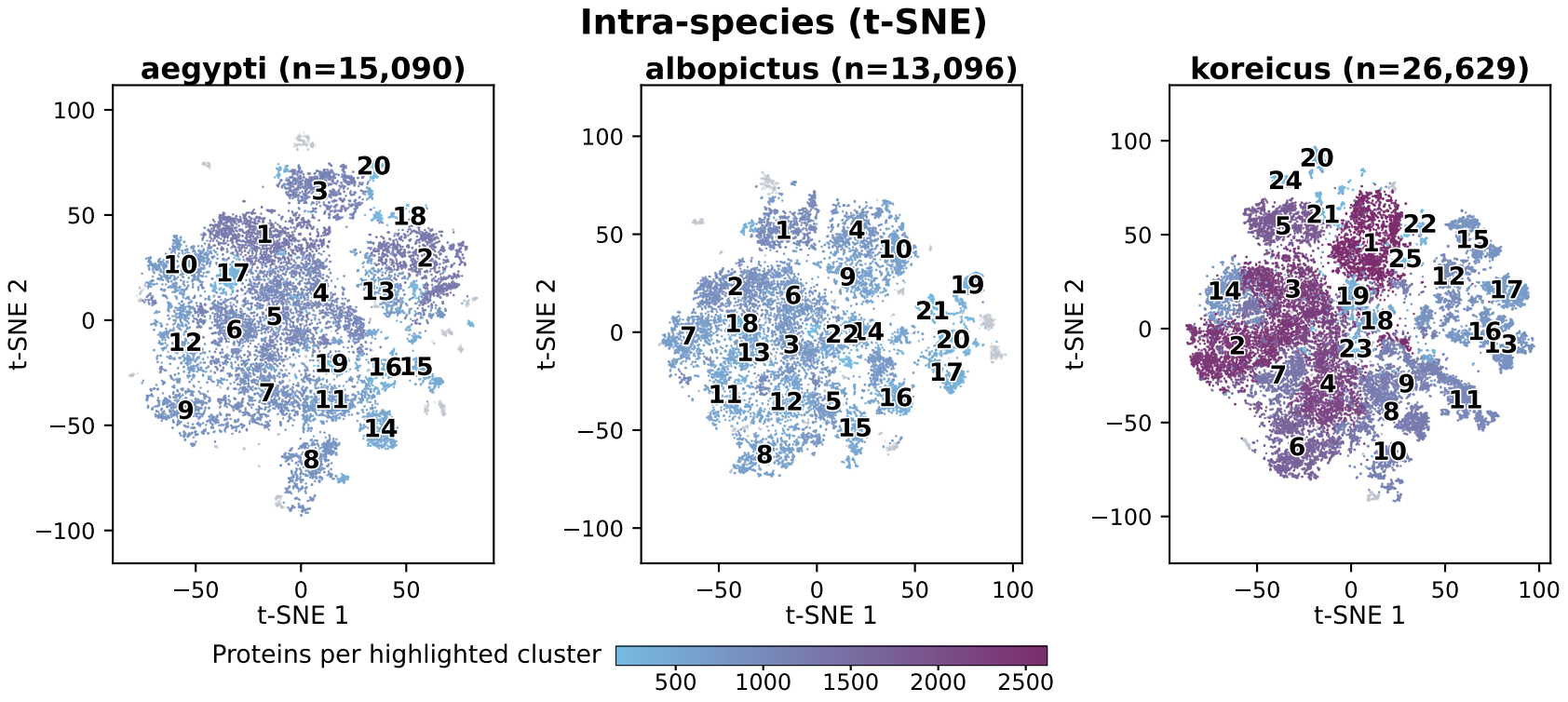
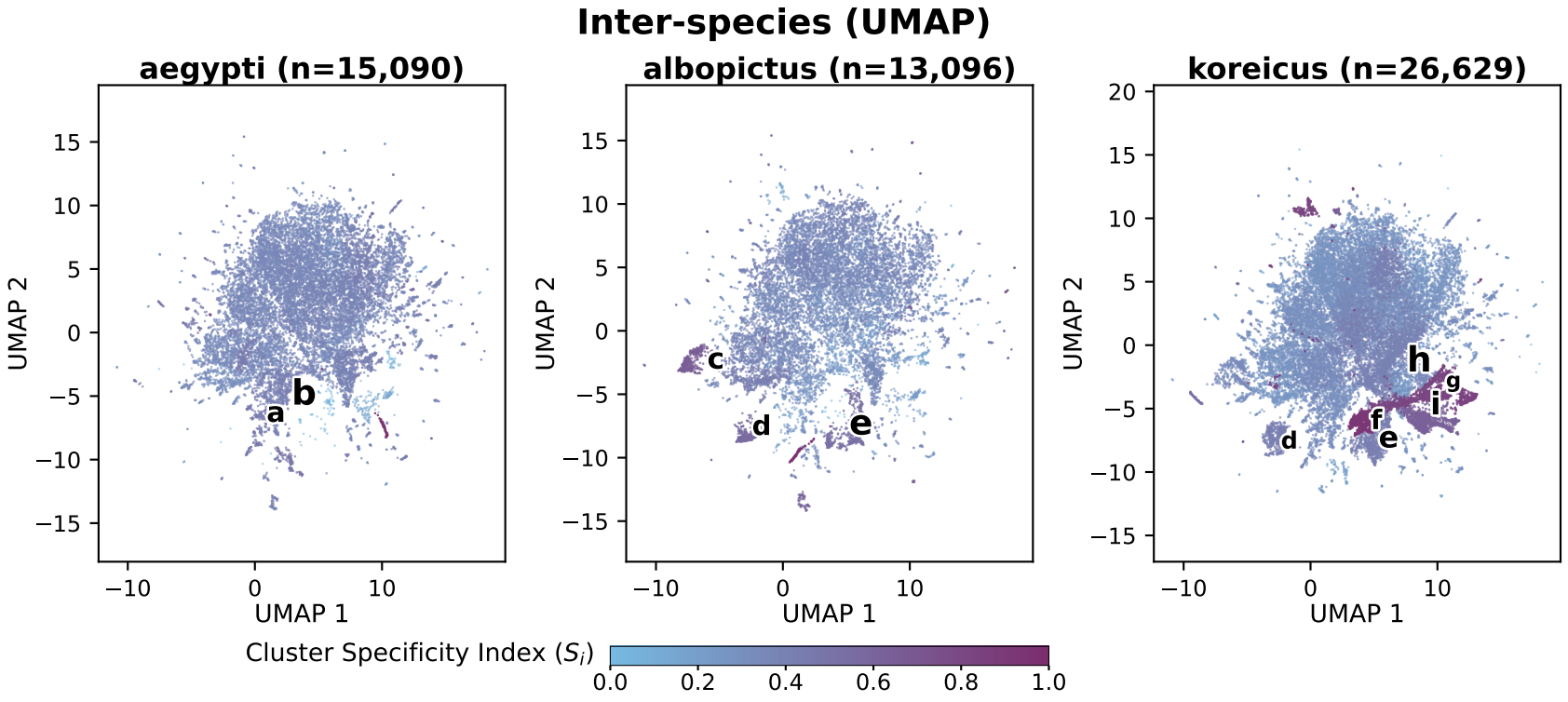
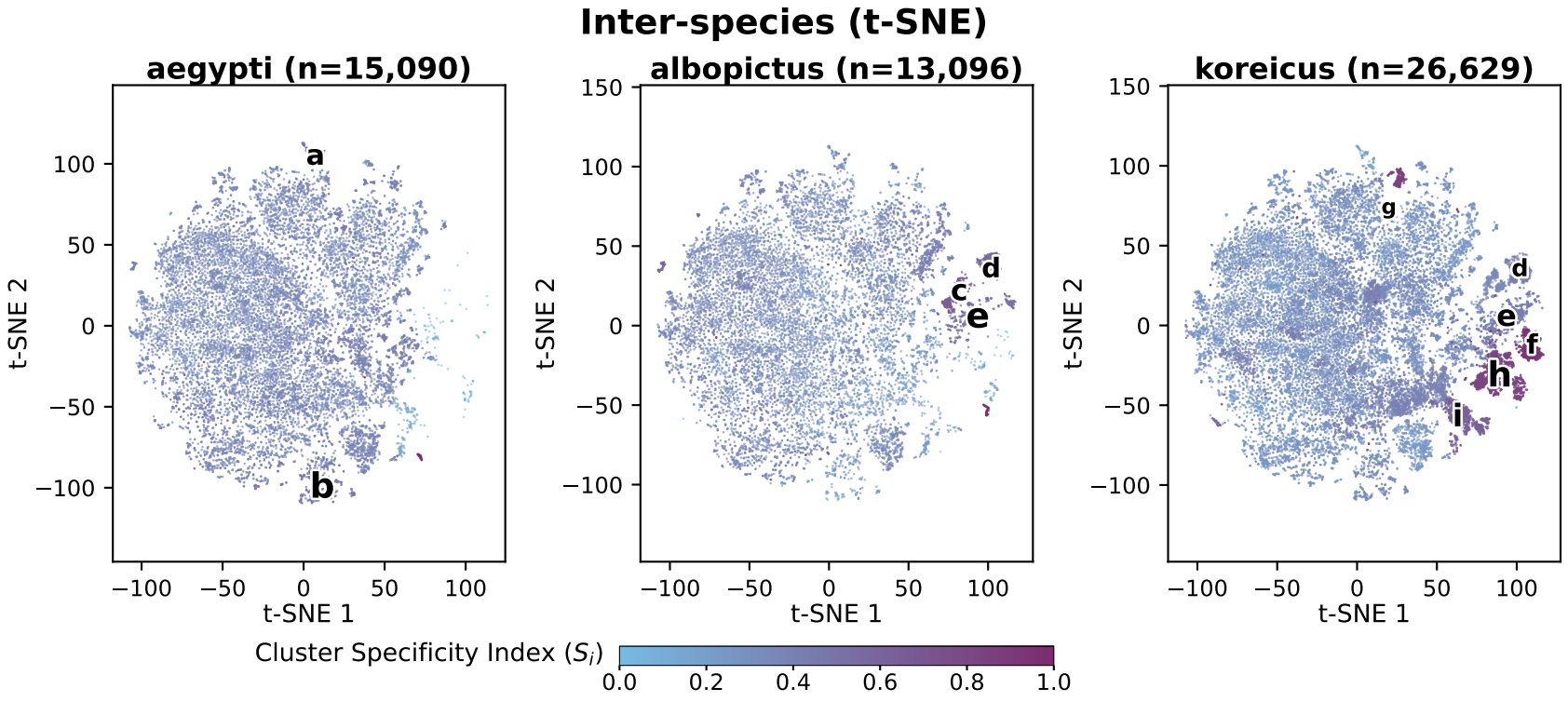
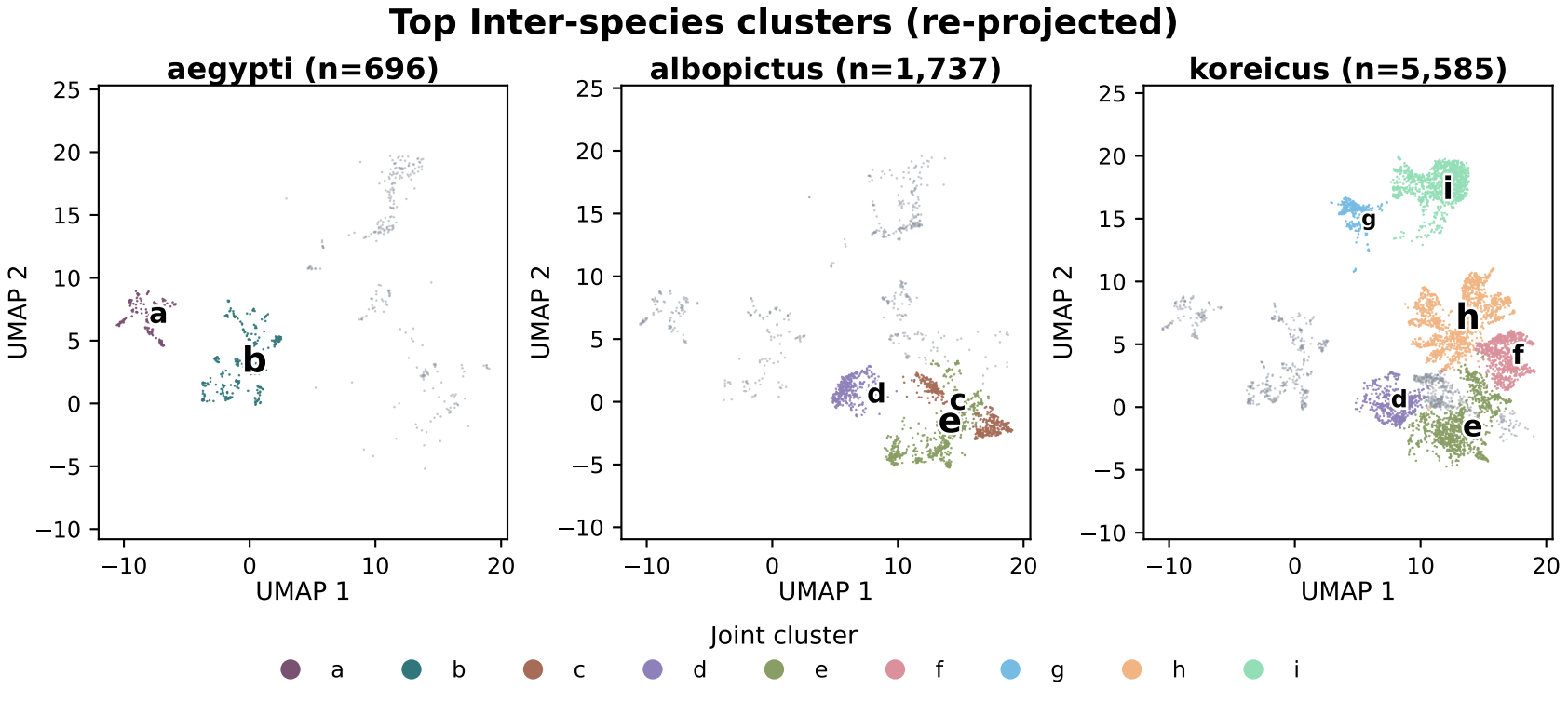
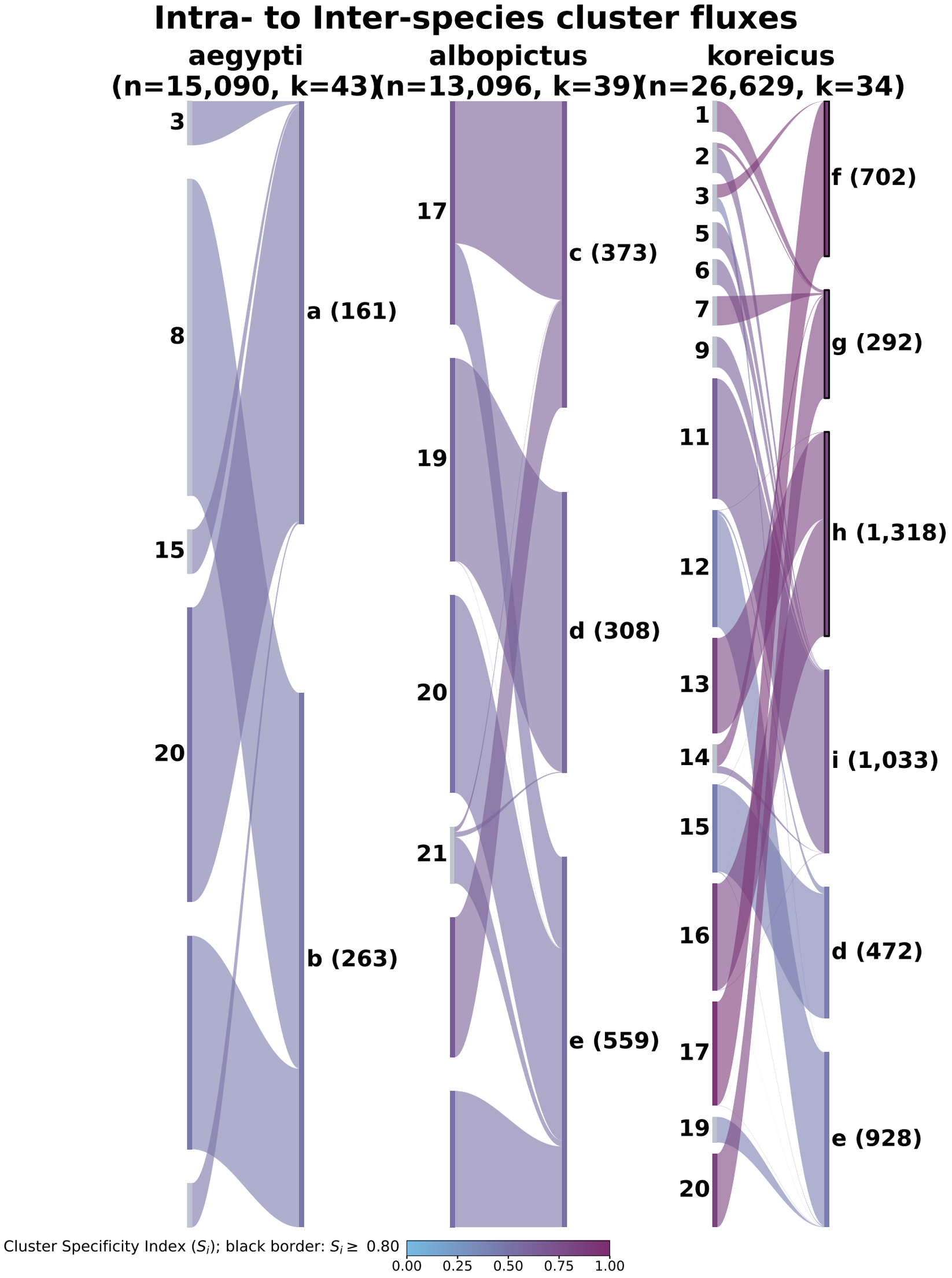

Intra-species (UMAP)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
UMAP 1
UMAP 2
Proteins per highlighted cluster
500
1000
1500
2000
2500

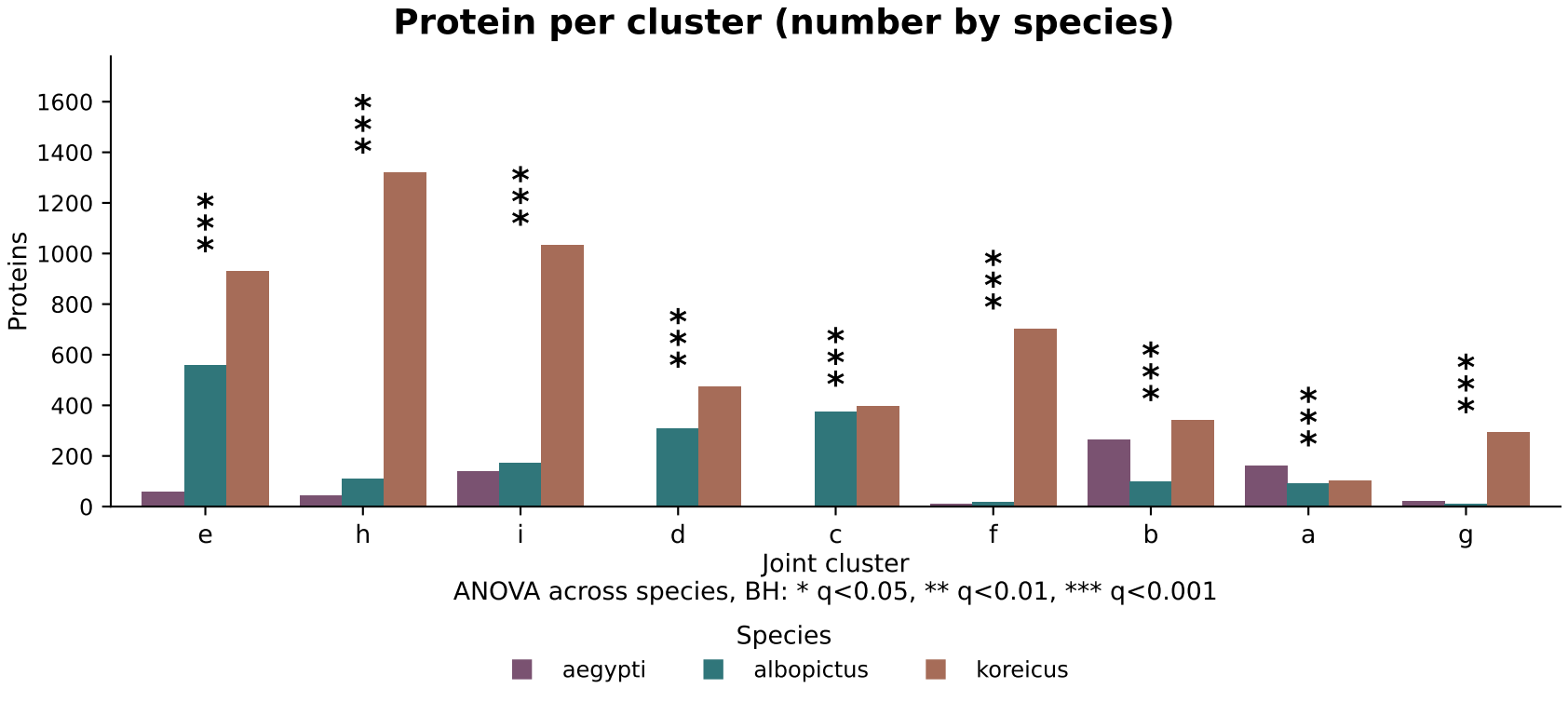

Intra-species (t-SNE)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
t-SNE 1
t-SNE 2
Proteins per highlighted cluster
500
1000
1500
2000
2500


Inter-species (UMAP)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
UMAP 1
UMAP 2
a
b
c
d
e
f
g
h
i
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Inter-species (t-SNE)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
t-SNE 1
t-SNE 2
a
b
c
d
e
f
g
h
i
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Top Inter-species clusters (re-projected)
aegypti (n=696)
albopictus (n=1,737)
koreicus (n=5,585)
UMAP 1
UMAP 2
Joint cluster
a
b
c
d
e
f
g
h
i


Intra- to Inter-species cluster fluxes
aegypti
(n=15,090, k=43)
albopictus
(n=13,096, k=39)
koreicus
(n=26,629, k=34)
3
8
15
20
a (161)
b (263)
17
19
20
21
c (373)
d (308)
e (559)
1
2
3
5
6
7
9
11
12
13
14
15
16
17
19
20
f (702)
g (292)
h (1,318)
i (1,033)
d (472)
e (928)
Cluster Specificity Index (S_i); black border: $S_i \geq 0.80$
0.00
0.25
0.50
0.75
1.00


Protein per cluster (number by species)
Proteins
1600
1400
1200
1000
800
600
400
200
0
****
****
****
****
****
****
****
****
e
h
i
d
c
f
b
a
g
Joint cluster
ANOVA across species, BH: * q<0.05, ** q<0.01, *** q<0.001
Species
aegypti
albopictus
koreicus